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Generating models for isoform-specific PKM-KIBRA interactions with BIFC, stabilization and AlphaFold 3

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Abstract

The truncated constitutive active form of protein kinase C (PKC) called protein kinase M (PKM) plays a role in long-term memory maintenance in vertebrate and invertebrate models. Previously we have shown that the *Aplysia* Kidney/Brain protein (KIBRA) stabilizes the atypical PKM Apl III, but not the classical PKM Apl I in *Aplysia* neurons. Expression of *Aplysia* KIBRA with changes in the proposed atypical PKM binding site does not stabilize PKM Apl III and erases forms of plasticity supported by PKM Apl III. Here, we have examined biomolecular fluorescence complementation (BIFC) between KIBRA variants and PKM Apl III in *Aplysia* neurons. These KIBRA variants include: the KIBRA with changes in the proposed atypical binding site noted above; a splicing variant that stabilizes PKM Apl I, but not PKM Apl III; and several mutations identified in mammalian WW and C2 domain containing protein 3 (WWC3, a member of the chordate-specific expansion of the KIBRA family) associated with cancer or neurodevelopmental disorders. Surprisingly, we find that some KIBRA variants show BIFC with PKM Apl III but do not stabilize PKM Apl III. We used models of protein-protein interactions (AlphaFold 3) to gain insights into the discrepancy between BIFC and stabilization of PKMs by KIBRA and KIBRA variants and suggest a model where stabilization is linked to stable inhibition of PKMs by KIBRA.

Keywords Persistent protein kinases, Synaptic plasticity, Memory retention, Protein complexes, Atypical protein kinase C, PKC

Introduction

A major challenge in our understanding of changes in synaptic strength underlying memory is how these changes are maintained for long periods of time, but at the same time can be erased by pharmacological agents. We have proposed that this challenge can be overcome

by specialized synapses where synaptic strength is maintained by persistent protein kinases underlying memory maintenance [39]. We have examined this hypothesis in *Aplysia* sensory-motor neuron synapses, where one can study long-term changes in synaptic strength related to long-term memory in a reduced system [21]. In these cultures, increases in synaptic strength associated with memory can be induced and maintained for days [17]. The increase in synaptic strength (and memory) in *Aplysia* are erased by inhibitors that target persistent protein kinases, in particular truncated forms of PKC that are termed PKMs [4, 15]. These inhibitors erase memory in many systems consistent with a model where

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mechanisms underlying long-term changes in synaptic strength, but not short-term changes, are highly conserved over evolution [52].

Two different forms of long-term facilitation, LTF can be induced in *Aplysia* sensory-motor neuron cultures, non-associative LTF (implicated in sensitization memory) induced by repeated applications of serotonin, and associative LTF (implicated in associative memory), induced by coupling firing of the sensory neuron and serotonin [15]. PKMs are formed by calpain cleavage in *Aplysia* and PKMs formed from classical PKCs (PKM Apl I) and atypical PKCs (PKM Apl III) play distinct roles in synaptic plasticity underlying memory formation [3, 8, 15]. Dominant negative forms of PKM can erase LTF, even if expressed two days after the increase in synaptic strength was induced, but distinct dominant negative PKMs block associative and non-associative LTF [15, 16]. Vertebrate PKM ζ binds to the adaptor protein KIBRA and altering three residues in the proposed PKM binding domain block this association [45]. Expression of *Aplysia* KIBRA with these three conserved residues changed (arginine 1317, serine 1319, arginine 1321 converted to alanines, KIBRA_{RSR-AAA}), erase LTF when injected into the sensory or motor neurons, matching the effects of dominant negative atypical PKM Apl III [16]. Drugs and peptides that interfere with KIBRA-PKM ζ association also reverse synaptic plasticity and memory in vertebrate models [44].

KIBRA stabilizes PKM ζ in vertebrates. Mice without KIBRA have significantly lower levels of PKM ζ , consistent with an endogenous role for KIBRA in stabilizing PKM ζ [45]. It is not clear if stabilization is simply due to association of PKMs with KIBRA. For example, KIBRA stabilizes the kinase LATS, but this is largely independent of its binding to LATS through its WW domains [49]. In contrast, KIBRA stabilization of PKM ζ in vertebrates has been linked to its association since the alteration mentioned above, KIBRA_{RSR-AAA}, shows neither stabilization nor co-immunoprecipitation [22, 45]. KIBRA also stabilized PKM Apl III in *Aplysia* sensory neurons but did not stabilize PKM Apl I [16]. Similar to vertebrates, KIBRA_{RSR-AAA} does not stabilize PKM Apl III in *Aplysia* neurons but surprisingly does stabilize PKM Apl I [10, 16]. There is an alternatively spliced insert in *Aplysia* KIBRA [10]. Like KIBRA_{RSR-AAA}, KIBRA_{splice} stabilizes PKM Apl I, but not PKM Apl III [10]. PKM Apl I is required in the postsynaptic neuron for maintenance of non-associative LTF [15].

KIBRA is a conserved adaptor protein implicated in cell polarity, the Hippo pathway and cell proliferation [11, 41, 51]. Atypical PKC is also implicated in these pathways and KIBRA and the atypical PKC are both participants in protein complexes implicated in these pathways [42, 51, 53]. KIBRA polymorphisms in humans are associated

with cognitive differences [2, 32, 33] and in neurons, KIBRA, also known as WWC1 and its other family members, WWC2 and WWC3 (formed by duplications during chordate evolution) localize to synapses and regulates both behavior and synaptic physiology underlying memory and behavior [6, 12, 22, 28, 29, 30, 43, 44, 45].

KIBRA variants show differential stabilization of PKM Apl I and PKM Apl III [10]. However, there has been no investigation of the relationship between interaction and stabilization in *Aplysia*. BIFC allows for measurements of interaction in a cellular environment [25, 26] and has been used to examine KIBRA-PKM ζ interactions in vertebrate cells [44, 45]. Here, we determine, for PKM Apl III, if changes in stabilization are correlated with changes in BIFC. Mutations in WWC3 near to the proposed PKM binding site are linked to neurodevelopmental disorders or cancer [18, 19, 40]. We also examined the effects of these mutations on stabilization of PKM Apl I and PKM Apl III and BIFC with PKM Apl III. Overall, we found that association measured by BIFC is not always correlated with stabilization. To gain insight into this, we have used AlphaFold 3 to model the KIBRA-PKM interaction and results from this model suggest that stabilization is linked to interactions where KIBRA stably inhibits PKM.

Materials and methods

Aplysia cell culture

Hermaphrodite *Aplysia californica* from Miami *Aplysia* Resource Facility (RSMAS) were anesthetized via injection of 50–60 mL of 400 mM isotonic MgCl₂ and pleuropedal ganglia were removed. All animal experiments were carried out in compliance with protocols approved by the Montreal Neurologic Institute animal committee (Protocol MNI-5784). Ganglia were digested at 19 °C in L15 media containing 10 mg/ml Dispase II (Roche Diagnostics) for 18–19 h. L15 medium (Sigma-Aldrich) was supplemented with 0.2 M NaCl, 26 mM MgSO₄, 35 mM dextrose, 27 mM MgCl₂, 4.7 mM KCl, 2 mM NaHCO₃, 9.7 mM CaCl₂, and 15 mM HEPES, with pH 7.4. Glass bottom culture dishes were coated with 0.05% poly-L-lysine for 1–2 h and washed with ddH₂O prior to use. Sensory neurons were isolated from pleural ganglia that were dissected from adult *Aplysia* (60–100 g). Neurons were cultured in 50% hemolymph/50% L15 media supplemented with L-glutamine. All experiments were performed at 19 °C.

Plasmid constructs and microinjection

All constructs were made in the pNEX3 vector [20]. mRFP-PKM Apl I, mRFP-PKM Apl III, KIBRA_{WT}, KIBRA_{RSR-AAA} and KIBRA_{splice} were previously described [10]. The enhanced cyan fluorescent protein (eCFP)-KIBRA fusion proteins were made using PCR to amplify eCFP with BspE1 and KpN sites and inserted into the

pNEX-KIBRA_{WT}, KIBRA_{RSR-AAA} or pNEX-KIBRA_{splice} plasmid cut with the same sites. Three GBLOCKS (Integrated DNA technology, IO, USA) were purchased with: the arginine at residue 1312 converted to glutamine; KIBRA_{R-Q}; the arginine at residue 1341 converted to histidine KIBRA_{R-H}; and the arginine at residue 1312 converted to a stop codon KIBRA_{stop}). These GBLOCKS were cut with Afe I and AflII and inserted into the CFP-KIBRA_{WT} or CFP KIBRA_{splice} cut with Afe I (partial) and Afl II. GBLOCKS for the long N-terminal part of mVENUS (1-154) and short C-terminal part of mVENUS (155-239) [24] were generated with restrictions enzymes SbfI and PpuMI and inserted at these sites in all pNEX-KIBRA variants or with restrictions enzymes Age I and Xho I to be inserted into pNEX-mRFP PKM Apl III. The YFP fragments were separated from PKM or KIBRA with a GGGGSGGGGS linker. All plasmids were confirmed by sequencing. For stabilization experiments, molar equivalent levels of plasmids encoding KIBRA variants and mRFP-PKMs were used. A solution containing the desired constructs (max 0.4 ug/ul of DNA in ultrapure water) with 0.2% fast green were microinjected into the nuclei of neurons using back-filled glass micropipettes. A short pressure pulse was delivered to release plasmid solution into nucleus. Neurons were incubated at 19 °C for 24 h to allow sufficient time for expression of injected plasmids prior to imaging.

Stabilization experiments

To measure stabilization by KIBRA, we express mRFP-tagged PKMs in the sensory neuron in the presence or absence of eCFP-KIBRA variants and measure the intensity of mRFP in processes (to avoid pigment granules in the cells soma that interfere with the measurement) normalized to the expression of eYFP (the plasmid encoding eYFP is co-injected with plasmid expressing mRFP-tagged PKMs) [10]. A blind observer chooses one region of interest (ROI) per neuron representing a sensory neuron neurite of intermediate thickness for the measurement. While some previous experiments used multiple ROIs, variability between different ROIs is low [5] and previous stabilization experiments used one ROI [10]. The level of expression is measured at 24 h after injection and low levels of plasmids are used to minimize toxicity [10].

Images were captured by an LSM 710 (Zeiss) laser confocal scanning microscope equipped with an Axiovert 100 inverted microscope (Zeiss) and a 40x, NA 1.4 objective. The eCFP, eYFP and mRFP, images were acquired sequentially. Fluorescence intensity (arbitrary units) was measured by the Microcomputer-Controlled Imaging Device (MCID) software package (Imaging Research). Background fluorescence from a neighboring region was subtracted from the fluorescence values measured.

BIFC experiments

To measure interaction by BIFC, we expressed three plasmids, two fusion proteins with non-fluorescent protein fragments, one long fragment (mVENUS 1-154) one short (mVENUS 155-239), and one encoding eCFP as a marker of how much plasmid is injected in sensory neurons. Fluorescence excitation and emission were achieved with an Xcite Expo arc, which delivers rich spectral excitation and uniform widefield fluorescence illumination to the sample plane. The images were captured through epifluorescence microscopy (widefield), and eCFP, eYFP and in some cases red images were acquired sequentially. Sensory neurons were imaged with a Zeiss Axio Observer D1, an EC Plan Neofluar 1.31 Numerical aperture 40x oil coupled lens, and a QuantEM: 512SCCM-CCD camera (Photometrics). The glass-bottom dishes were filled with 1X PBS instead of fresh L-15-based media to avoid background fluorescence. Rapid image acquisition was achieved with a Zeiss SVB-1 microscope signal distribution box and Axiovision 4.8 software using the fast acquisition application (Zeiss). Initiation of the Axiovision image acquisition triggers the recording of three channels (DIC, eCFP, eYFP) or in addition Red (for AlexaFluor647 used in antibody staining). Similar, to stabilization experiments, A blind observer chooses one ROI per neuron representing a sensory neuron neurite of intermediate thickness for the measurement. We measured the intensity of eYFP in processes and normalized it to the intensity of eCFP. For expression levels we measured the intensity of Red staining and normalized it to the intensity of eCFP. For all measurements, a region next to the neurite was chosen for background subtraction.

In some experiments where an antibody to KIBRA was used, cultured cells were fixed with 4% PFA + 30% sucrose in PBS for 30 min and washed with PBS. Fixed cells were permeabilized with 0.1% Triton X-100 with 30% sucrose in PBS for 10 min, and free aldehydes were quenched (50mM ammonium chloride for 10 min, followed by four washes. Cells were incubated with 10% normal goat serum (Sigma Aldrich) and 0.5% Triton X-100 in PBS for 30 min to block nonspecific antibody binding. Cells were then treated with KIBRA Antibody [16] in blocking solution (1:500) for 1 h, followed by five PBS washes (10 min each). Cells were then incubated with AlexaFluor-647-conjugated donkey anti-rabbit antibody in blocking solution (1:500; Invitrogen, REF A315773) for 24 h at 4 °C in darkness and washed with PBS as described above. Imaging was the same as above.

Quantification and statistics

In all experiments, N represents one sensory neuron. An independent experiment is defined when sensory neurons are obtained at the same time from the same animals and conditions are done in parallel. Sensory neurons

from multiple animals are pooled before plating for each independent experiment.

Statistical methods were not used to predetermine sample size. For stabilization experiments, previous experiments [10, 16] indicated that a minimum of three independent experiments and an N of greater than 15 sensory neurons was sufficient to show an effect of KIBRA on stabilization. Since there were multiple constructs and not all constructs were included in each experiment, the number of experiments and sensory neurons were always greater than this minimum. For BIFC experiments, we had not established the variability of this measurement and planned 4 independent experiments for each construct, each including positive (KIBRA_{WT} Long/PKM Apl III short) and negative (PKM Apl III long/PKM Apl III short) controls. Similar to stabilization experiments, since there were multiple constructs and not all were included in each experiment, the number of experiments were greater than this minimum.

For each independent experiment, measures (including the control group) were normalized to the average of the control group (pNEX alone in stabilization experiments, KIBRA long/PKM short in BIFC experiment, KIBRA_{WT} in expression studies). There was no normalization for the initial characterization of BIFC constructs. All quantifications were performed blind to the experimental group.

No test for outliers was done. Shapiro-Wilk tests for normality were done for all groups and for all experiments, at least one group did not pass the normality test ($p < 0.05$). Thus, for all experiments, the non-parametric Kruskal-Wallis test was used to determine if there were significant differences between the groups. Pre-planned Dunns Post-Hoc tests were planned to test for significant stabilization or BIFC compared to the negative control (pNEX for stabilization, PKM Apl III long/PKM Apl III short for BIFC). All statistical tests were done with GraphPad Prism version 9.

AlphaFold3 simulations

Structural models were generated with AlphaFold 3 [1] on the AlphaFold 3 server (<https://alphafoldserver.com/>). All described interactions are predicted and have not been experimentally validated. The terms 'interaction', 'binding', and 'contact' in the context of AlphaFold 3 results refer to predicted structural proximity, not confirmed molecular interactions. These models were generated with default template settings (PDB templates with default cut-off date (29/09/2021)). The proteins used were PKM Apl III (NP_01191516.1 starting at M238 with PTMs as pT179 and pT331, PKM Apl I (NP_00191415.1 starting at Y300 preceded by 7 linker residues (SGLRSRG) with PTMs at pT186, pT328, and pS347), PKM zeta (NP_001229803.1 starting at M80 with PTMs at pT227

and pT377), KIBRA_{WT} (XP_35825059.1), KIBRA_{splice} (XP_035825058.1), Human KIBRA (NO_001155133.1), Human WWC2 (NP_0013977931.1), Human WWC3 (N_056506.3), Par3 (NP_001171720.1). For KIBRA mutants (KIBRA_{RSR-AAA}, KIBRA_{R-Q}, KIBRA_{R-H} and KIBRA_{stop}) all models used the same sequence as KIBRA_{WT} with the changes described in the text. All models also contained ATP and 2 Mg²⁺ ions. Unless otherwise stated, all queries were run five times to generate 25 models, to sample differences that might arise from the multiple sequence alignment process and the random model generation start sites. All models were subsequently analyzed with Molprobit [47]; hydrogens were added to models with Reduce (NOFLIP flag) with electron-cloud x-H bond lengths to preserve predicted side chain orientations, then the models were scanned for residue atomic clash above 0.4 Å and Ramachandran probability scores. Mean clash per residue was calculated as the mean significant clash ($> 0.4 \text{ Å}$) per residue over the stated region from individual residue clash matrices. Representative models were selected as the model with no severe clashes ($> 0.9 \text{ Å}$), 100% favored Ramachandran scores, and the lowest overall mean clash per residue over the canonical substrate interface (KIBRA_{WT} 1310–1323). AlphaFold generated predicted Local Distance Difference Test (pLDDT) is reported as the mean $\pm$ pooled SD for the stated region and AlphaFold generated interface predicted Template Modelling (ipTM) reported is whole model ipTM. AlphaFold 3 output data from sequence queries were subsequently analysed and visualized with Pymol Molecular Graphics System [36]. For all models with two or more proteins, for each residue the minimum interchain Predicted Alignment Error (interchain PAE or iPAE) was found to identify regions of potential protein-protein interaction. The lower the minimum iPAE (Å) for a residue is, the more confidence in the position of that residue with respect to a residue in the other protein, indicative of participation in a protein-protein interaction. The mean minimum iPAE averages the data over all 25 models to report the relative consistency of the interactions observed in the individual models. The minimum iPAE data highlights focused predicted protein-protein interactions, where the measure directly reflects changes in predicted interaction strength and the specific predicted location of the interaction. However, not all likely protein-protein interactions will be modelled with low iPAE at the interacting residues. If an interaction occurs through a domain that allows alternative residue interactions, iPAE may be relatively low, as specific residues are modeled interacting with a variety of residues in the other protein. To identify such potential protein-protein interactions involving domains we scanned all models with Pymol for interactions, including hydrophobic based on residue identity and $< 4 \text{ Å}$ sidechain proximity,

hydrogen bonds based on N/O proximity ($< 3.5\text{\AA}$) and donor/acceptor angle ($> 120^\circ$, yellow dashed lines in models), salt-bridges through residue identity and proximity ($< 4.5\text{\AA}$, green dashed lines in models are $< 4.0\text{\AA}$), and pi- interactions through residue identity, proximity and angle to ring normal ($< 4.5\text{\AA}$; $< 45^\circ$ cation-pi, purple dashed lines) that may underlie protein-protein interactions. Domains involved in potential protein-protein interactions show regular interactions with the same target domains in the other protein over the twenty-five models. Error bars on mean minimum iPAE charts are standard error of the mean.

Results

KIBRA and PKM Apl III show BIFC, even when they do not show stabilization

In BIFC, association of two proteins, one containing an N-terminal fragment (long, residues 1–154) of eYFP and the other containing a C-terminal fragment (short, residues 155–238) of eYFP allows formation of eYFP when

the two proteins associate [23, 24]. We quantified the amount of yellow fluorescence in sensory neuron processes as pigment granules in the cell soma of sensory neurons interfere with measurements of fluorescence in the soma [10]. To control for the level of expression, we co-injected a plasmid encoding eCFP and used the eYFP/eCFP ratio as a measure of BIFC. KIBRA and PKMs were injected with an equal molar ratio of their encoding plasmids. We used similar placement of the two YFP fragments at the N-terminal of KIBRA and PKM as used for BIFC in mammalian cells [44, 45]. We compared the amount of yellow fluorescence to when the two fragments of eYFP were both on KIBRA or both on PKM and found significantly more YFP when the fragments were on KIBRA and PKM, demonstrating an interaction between KIBRA and PKM in sensory neurons that could be monitored by BIFC (Fig. 1).

We examined BIFC for two variants of KIBRA that led to no significant stabilization of PKM Apl III, KIBRA_{RSR-AAA} [16], and KIBRA_{splice} [10], and compared

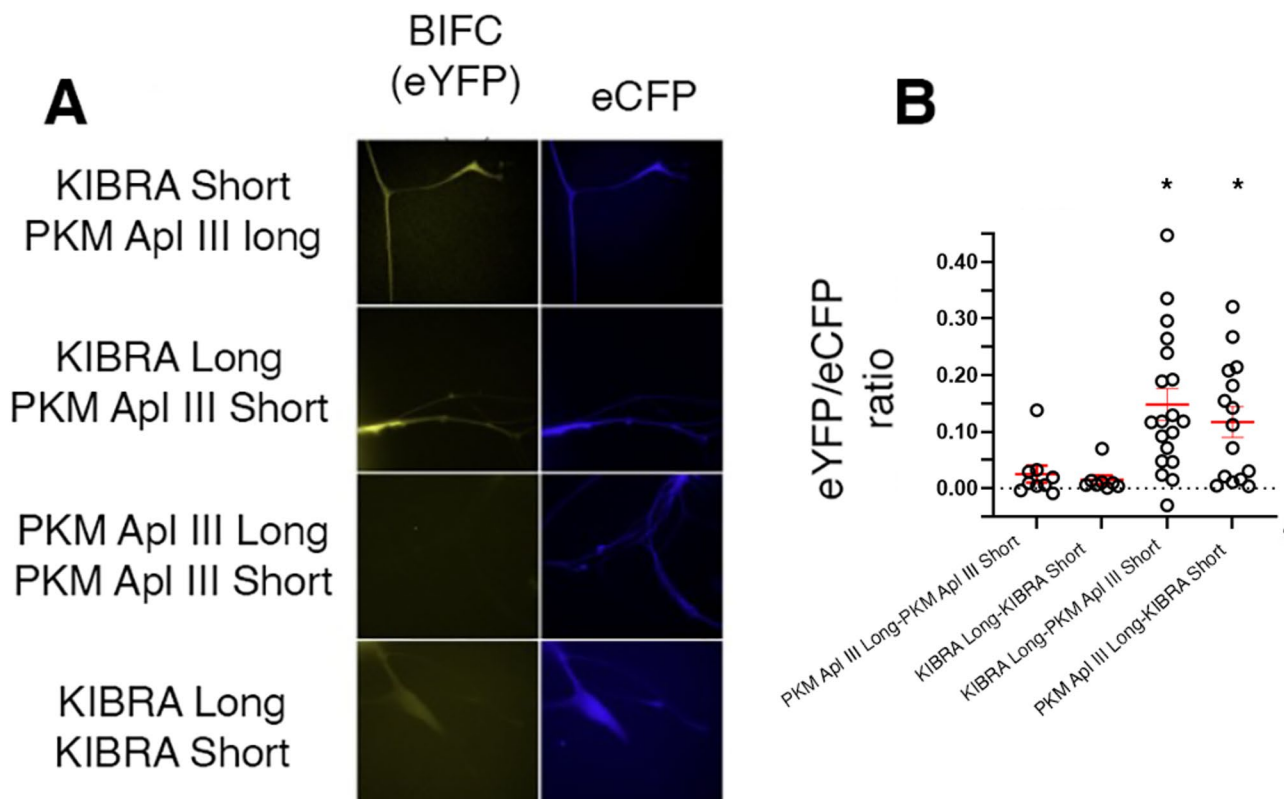


Fig. 1 BIFC between KIBRA and PKM Apl III in *Aplysia* sensory neurons. **A** BIFC construct expression where long refers to the N-terminal of YFP (1–154) and short refers to the C-terminal of YFP (155–238). Representative examples of expression of eCFP (control for injection), and eYFP (BIFC) are seen for the four groups (Kibra Short/PKM Long, KIBRA Long/PKM Apl III Short, KIBRA Long/ KIBRA Short, PKM Apl III Long/ PKM Short). **B** All individual points (black circles) and the average and SEM (red) of the unnormalized ratio of eYFP/eCFP is shown. Kruskal-Wallis test (51,4) = 17.52, $p < 0.001$. Dunn's multiple comparison test showed that PKM Apl III long/PKM Apl III short was significantly different than KIBRA long/PKM Apl III short ($p < 0.05$, *) and PKM Apl III long/Kibra short ($p < 0.05$, *) and KIBRA long/KIBRA shot is significantly different than KIBRA long/PKM short ($p < 0.01$). Results from four independent BIFC experiments with the following number of sensory neurons: PKM Long/PKM Short 9, KIBRA Long/KIBRA Short 8, KIBRA Long/PKM Apl III Short 19, KIBRA Short/PKM Apl III Long 18

results with KIBRA_{WT}. We were surprised to find that in both cases, modified KIBRAs showed strong significant BIFC (Fig. 2). The levels of expressed KIBRA were comparable as monitored by an anti-KIBRA antibody (Fig. 2). Thus, association of KIBRA and PKM measured by BIFC does not always lead to stabilization of PKM.

AlphaFold 3 modeling of KIBRA-PKM interactions reveals alternative substrate peptides in KIBRA

To gain more insight into how KIBRA and PKM associate and why association may not lead to stabilization, we used AlphaFold 3 [1] to examine the predicted interaction between forms of KIBRA (KIBRA_{WT}, KIBRA_{splice}, KIBRA_{RSR-AAA}) and PKM Apl III. We also examined the predicted interaction of these forms of KIBRA with PKM Apl I since both KIBRA_{splice} and KIBRA_{RSR-AAA}, but not KIBRA_{WT} stabilize PKM Apl I [10]. The most confident interaction predicted by AlphaFold 3 for all KIBRA-PKM interactions is a substrate-like interaction where a PKM-interacting peptide from KIBRA binds in the substrate binding site of PKMs (Fig. 3A). For PKM Apl I, AlphaFold 3 has two possible paths for this peptide, equally represented in the AlphaFold 3 models (Fig. 3A). One path was similar to the one modelled for PKM Apl III and the other is discussed below. The AlphaFold 3 predicted interaction path for the KIBRA substrate peptide

with PKM Apl III is very similar to the crystal structure made previously examining the interaction between a peptide sequence in PAR3, that is similar to the KIBRA peptide, and atypical PKC iota [38]. For the PAR3 peptide, interaction sequences surrounding the phosphorylatable serine in the PAR3 peptide leads to inhibition of phosphorylation by PKC iota [38]. These sequences comprise an affinity arm (phenylalanine, X, arginine; FxR) on the N-terminal side of the substrate serine residue and an inhibitory interaction on the C-terminal side of the substrate residue [38]. KIBRA also has the FxR sequence on the N-terminal affinity arm and homology in the C-terminal inhibitory region (Fig. 3A). The AlphaFold 3 model for Human PAR3 with PKM Apl III predicts a similar interaction for PAR3 as seen with PKC iota [38]. In particular, the affinity arm FxR is predicted to have very similar paths with the same interchain interaction residues predicted for PAR3 and KIBRA (Fig. 3B).

KIBRA stabilizes PKM Apl III, but not PKM Apl I, but AlphaFold 3 predicts that the KIBRA peptide can fit into the substrate binding site of both PKMs (Fig. 3Bii). However, there is an alternative predicted binding of PKM Apl I (Fig. 3Biii) where the FxR sequence of KIBRA may interact with what we have termed the handle region of PKM Apl I, a region including an α -helix (helix 5 or helix G in the catalytic domain structure of PKCs) that

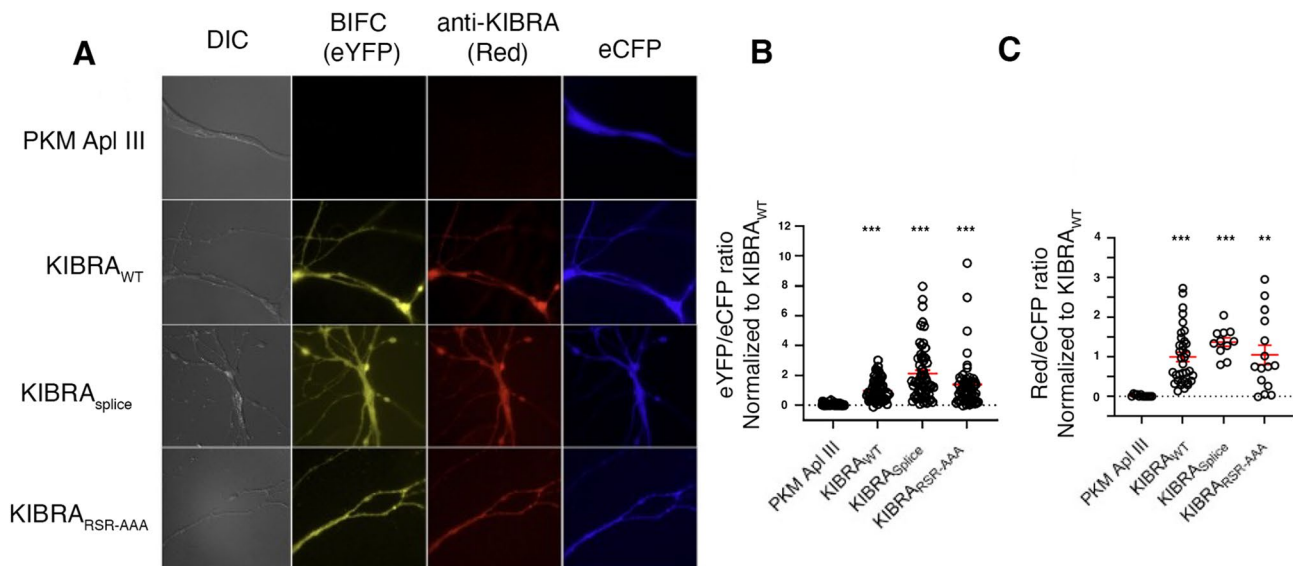


Fig. 2 Comparing BIFC between KIBRA variants and PKM Apl III **A** Representative examples of expression of DIC (Differential Interference Contrast), eYFP (representing BIFC signal), red (Kibra Antibody), and eCFP (control for injection) are seen for the four groups all expressing PKM Apl III short and expressing one of PKM Apl III Long, KIBRA_{WT} Long, KIBRA_{splice} Long, KIBRA_{RSR-AAA} Long. **B** All individual points (black circles) and the average and SEM (red) of the normalized eYFP/eCFP ratio (see Methods) is shown. Kruskal-Wallis test (221,4) = 105.9, $p < 0.0001$. Dunn's multiple comparison test showed that all groups were different than PKM Apl III Long ($p < 0.001$, ***). The N was the number of sensory neurons quantified (PKM Long 46, KIBRA_{WT} Long 64, KIBRA_{splice} Long 59, KIBRA_{RSR-AAA} Long 52) and results are from 8 independent experiments. **C** All individual points (black circles) and the average and SEM (red) of the normalized Red/eCFP ratio is shown. Kruskal-Wallis test (73,4) = 30.32, $p < 0.0001$. Dunn's multiple comparison tests showed that all groups are significantly different than PKM Apl III (only endogenous KIBRA) with $p < 0.001$ (***), or for KIBRA_{RSR-AAA}, $p < 0.01$ (**). There were no differences between the KIBRA expressing groups ($p > 0.1$). Numbers are different from the BIFC experiments, as antibody staining was not done for all experiments. The N was the number of sensory neurons quantified PKM Apl III Long 12, KIBRA_{WT} 35, KIBRA_{RSR-AAA} 11, KIBRA_{splice} 15) from 4 independent experiments

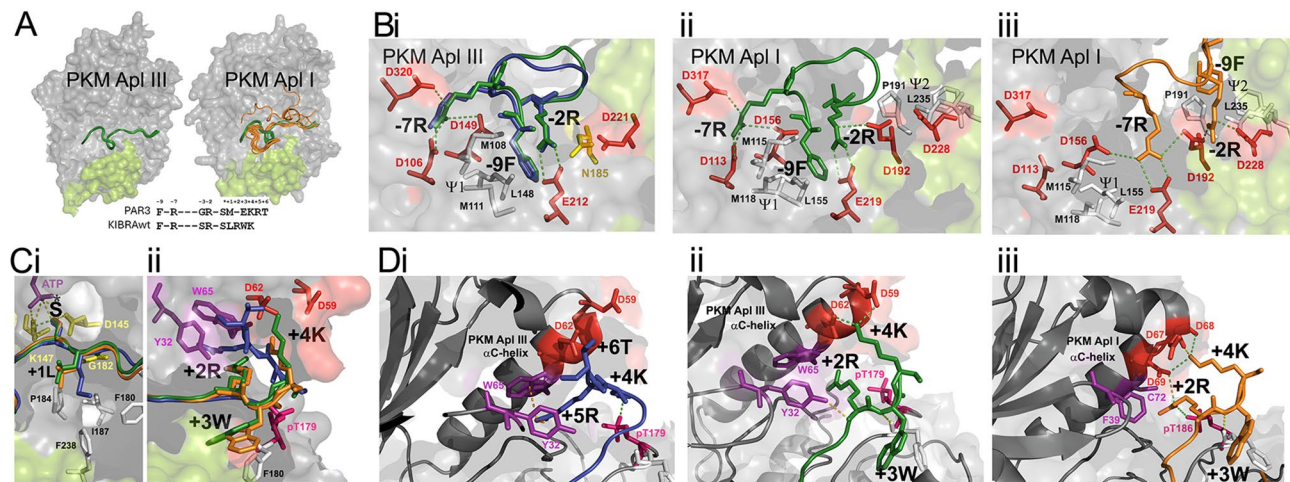


Fig. 3 AlphaFold 3 models of Aplysia KIBRA_{WT} or Human PAR3 substrate peptides with PKM Apl III and PKM Apl I. **A** Twenty-five models overlaid and aligned by the PKM, showing a consistent KIBRA_{WT} substrate (green) path through PKM Apl III (model ipTM range 0.87–0.88, all KIBRA_{WT} structures hidden except the canonical PKM substrate (1310–1323) (left). Half of the models show a similar substrate path through PKM Apl I (green, 13 of 25 models, with ipTM range 0.8–0.86), however the other half of the models with PKM Apl I show an alternate path for this peptide with ipTM range 0.65–0.68 (9 models) or low confident paths < 0.6 ipTM (3 models) (orange), often interacting with the handle region (olive) of PKM Apl I. Sequence alignment of the Human Par3 substrate and the Aplysia KIBRA substrate residues consistently modelled to be involved in interactions with PKM Apl III numbered relative to the potential phosphorylation residue, P-0 marked as *. **B** (i). Human PAR3 (blue) and Aplysia KIBRA_{WT} (green) substrate ‘affinity arm’ side chains of -9 F in the hydrophobic pocket (Ψ1), and -7 R and -2 R in separate acidic pockets of PKM Apl III (PKM surface in grey/yellow, acidic side chains of PKM Apl III in red, hydrophobic side chains of PKM Apl III in white). (ii) The substrate modeled with PKM Apl I, has a similar path (ii, green) or an alternative path (iii, orange) involving -9 F interacting with the hydrophobic pocket in the handle (Ψ2) while -7 R interacts with the same pocket as -2R. D192 in PKM Apl I permits this alternative interaction with both -7 R and -2R, while the orthologous N185 in PKM Apl III supports only the -2R with a -3 S hydrogen bond (yellow in Bi). **C** (i) Substrates interacting at the target serine (S) with PKM catalytic loop residues D145/K147, and the +1 hydrophobic residues sit in a hydrophobic pocket and H-bonds to G182 (PKM side chains predicted to form hydrogen bonds in yellow, hydrophobic interactions in white, ATP in violet, numbering corresponds to PKM Apl III). C(ii) Substrates ‘inhibitory arm’ interacting with three regions in PKM Apl III at Y32&W65 (magenta), at pT179 & F180 (hotpink, white), and at D59 & D62 (red), KIBRA substrate residues +2R, +3 W and +4 K labeled. **D** (i) Par3 interactions with PKM Apl III with residues orthologous to those determined for PKC iota and PAR3 [38], +4 K interacting with pT179, +5R with Y32&W65, and +6T with D62. (ii) KIBRA +2R models suggest a cation-π interaction to W65 and a salt bridge to pT179, +3 W interacts with the +1 hydrophobic pocket, and +4 K with the acidic patch at the start of the αC helix, D62 and D59. (iii) The KIBRA substrate inhibitory arm interactions in models with PKM Apl I have a similar +3 W hydrophobic interaction, while +2R is predicted to interact with the primed residue pT186 and D69, while +4 K interacts with D67 and D68 in the αC helix. Representative single models selected based on the absence of severe clashes (> 0.9Å) and Ramachandran plot outliers, and with the lowest mean clash per residue at the substrate interface (KIBRA_{WT} 1310–1323, Human Par3 758–773) as measured with Molprobit [47]; the selected models had the following for substrate interface mean clash per residue, substrate interface pLDDT, and ipTM for Human Par3 in PKM Apl III 0.107Å, 79.69, 0.88. KIBRA_{WT} in PKM Apl III 0.092Å, 73.54, 0.88. KIBRA_{WT} with PKM Apl I green path 0.267Å, 73.06, 0.85. KIBRA_{WT} with PKM Apl I orange alt path 0.434Å, 73.06, 0.65

is important for differential stabilization of PKM Apl I and PKM Apl III [10]. The mean ipTM, an AlphaFold generated metric of interface confidence, is reduced for this alternate substrate path from 0.88 ± 0.01 , 0.85 ± 0.01 for KIBRA_{WT} with PKM Apl III and PKM Apl I respectively, to 0.64 ± 0.09 for the alternate substrate path in KIBRA_{WT} and PKM Apl I models (means $\pm$ SD from 25 models for PKM Apl III and 13 models for Par-3 like path, 12 models for alternate path for PKM Apl I). The alternate substrate path was modelled with AlphaFold runs over all five models and with a mix of the Par-3 like path and the alternate path, indicating that modelling of the alternate substrate path is not solely determined by the initial multiple sequence alignments, the start point of model generation, or the recycling process.

The phosphorylatable serine and +1 hydrophobic residue are modelled in similar interactions for both PAR3 and KIBRA_{WT} with PKM Apl III and with the

corresponding residues of PKM Apl I with KIBRA_{WT} (Fig. 3Ci). However, while the AlphaFold 3 prediction of the interactions of the inhibitory arm of the PAR3 peptide with PKM Apl III are similar to what was predicted for PKM iota (Fig. 3Di), the predicted interactions for the KIBRA peptide with PKM Apl III and for PKM Apl I show differences (Fig. 3Dii and Diii).

AlphaFold 3 also predicts substrate-like binding of alternative peptides. These two alternative sequences, which we call substrate peptide 1 (SP1) and substrate peptide 2 (SP2) directly precede the previously identified atypical PKC binding site that we now call substrate peptide 3 (SP3) (Fig. 4A). These regions are highly conserved in KIBRA ([10] Fig. 4A), but the AlphaFold 3 prediction for all substrate peptides is unstructured in KIBRA alone (mean of five models of KIBRA_{WT} alone SP1 to SP3 (aa1271-1323) pLDDT = 35.92. Interestingly,

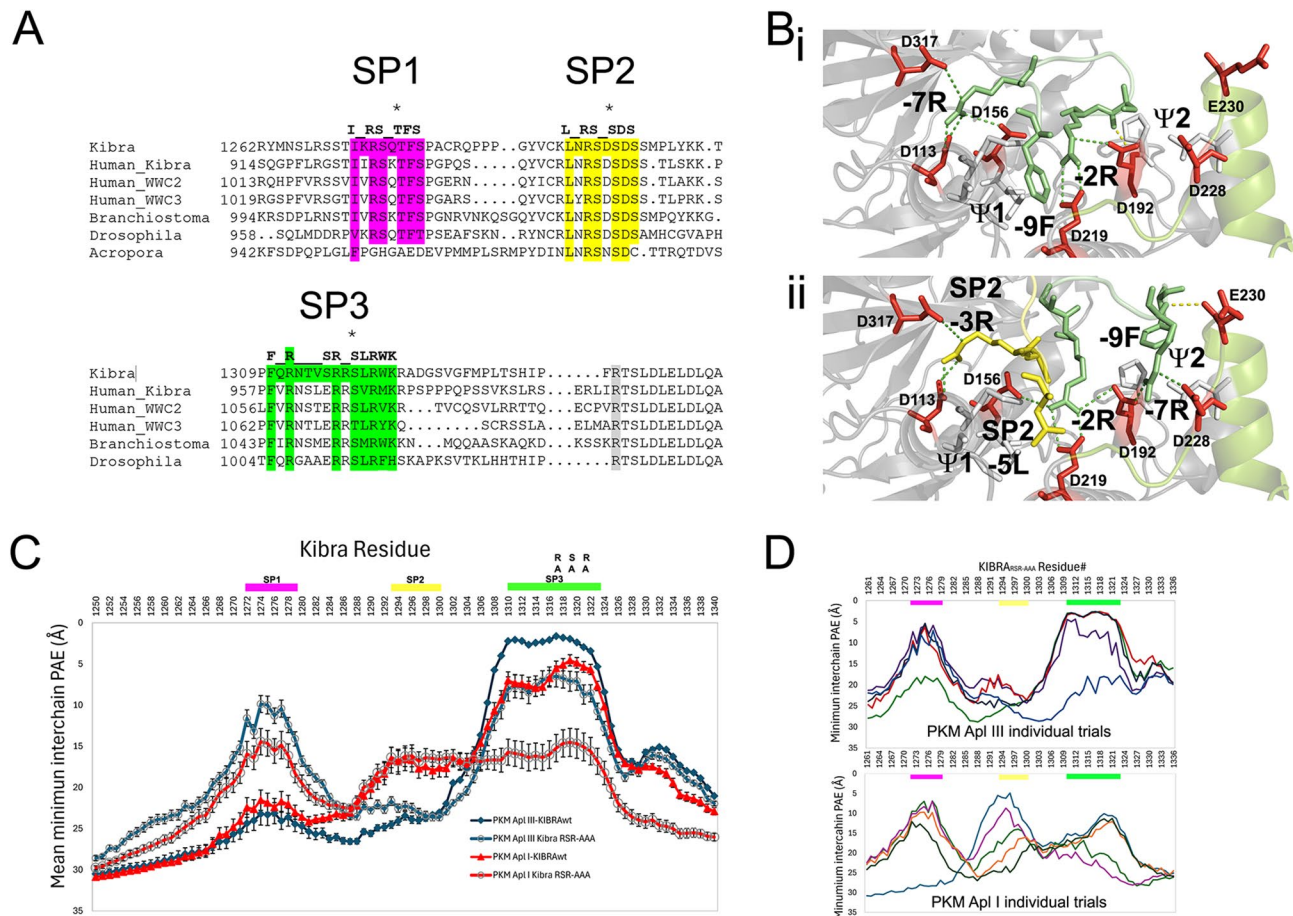


Fig. 4 Substrate peptide sequence interactions with PKM Apl III and PKM Apl I. **A** Sequence alignments of KIBRA molecules from Aplysia, Branchiostoma, Drosophila, Acropora, and the three Human isoforms. Bold letters above alignments show residues that can participate in substrate binding in AlphaFold models, with the canonical substrate in green (SP3) and the alternate substrate peptides in magenta (SP1) and yellow (SP2). The R-Q mutation and the KIBRAstop site, R1312 marked in green, and the R-H mutation, R1341 in grey. **B** Representative models of KIBRA_{WT} and PKM Apl I interaction from Fig. 3 with alternate substrate peptides in the -7R acidic pocket with -7R alternate binding to the -2R pocket. Bi) -7R and -2R of the KIBRA substrate (SP3) with PKM Apl I can interact with similar residues as PKM Apl III, with -7R and -2R forming salt bridges in separate pockets of acidic side chains (red). (ii) Alternative binding of -7R KIBRA SP3 peptide (green) to the PKM Apl I handle through D192& D228 allows alternate substrate SP1 or SP2 to interact with the former -7R pocket formed by D113, D156, D317 (example is of SP2 -3R in yellow). The -9 F of SP3 is predicted to interact with an alternative hydrophobic pocket (Ψ2) in the PKM Apl I handle region and the -5 hydrophobic residue of either SP1 or SP2 interacts with the former -9 F hydrophobic pocket (Ψ1,-5 L of SP2 shown here). Of the 25 models, 12 have the SP3 substrate in the alternate path, involving SP2 (9) or SP1 (3). The model iPTM is reduced from a mean of 0.85 (range 0.8–0.86) for twelve models with the canonical path, to 0.64 for alternate paths (range 0.47–0.68). Acidic sidechains in red, hydrophobic in white. **C** Mean minimum interchain PAE for the substrate binding region of KIBRA for PKM Apl III or PKM Apl I modeled with either KIBRA_{WT} or KIBRA_{RSR-AAA}. **D** Minimum interchain PAE of single models with KIBRA_{RSR-AAA} with PKM Apl III (top) or PKM Apl I (bottom). Individual models chosen to represent the variation in model minimum iPAE variation for the substrate region over the twenty-five models to represent the full data set and reduce clutter

in the alternative path for the KIBRA peptide in PKM Apl I where the FxR may interact with the handle domain, the residues in the kinase domain that were predicted to interact with FxR are now predicted to interact with the comparable regions of SP1 or SP2 (Fig. 4B). To compare these predicted interactions we used the minimum interchain predicted alignment error (iPAE) calculation from AlphaFold 3 that measures the confidence in residue proximity to all other residues. We use the mean of 25 models (see Methods) to calculate the mean minimum iPAE for each residue which gives a measure of potential

interaction [1]. PKM Apl III had lower iPAE (higher predicted interaction stability) in the SP3 region than PKM Apl I and higher iPAE (lower predicted interaction stability) with SP1 and SP2 than PKM Apl I (Fig. 4C). These potential interactions change with KIBRA_{RSR-AAA}. This dramatically shifts the predicted PKM interactions to SP1 from SP3 for both PKM Apl I and PKM Apl III (Fig. 4C). While the mean iPAE is useful, it can be misleading as it averages across quite distinct models. To illustrate this point, we plot 5 individual iPAE calculations in Fig. 4E to show this variability. Thus, while some models still show

strong predicted interaction of SP3 even in the absence of the conserved RSR sequence, a substantial number of models show higher iPAE for SP3 and lower iPAE for either SP1 or SP2 lowering the mean iPAE of SP3 (Fig. 4D). The ability of the other substrate peptides to replace SP3 may explain why PKM Apl III still shows BIFC with KIBRA_{RSR-AAA}.

We used Molprobit [47] to choose the best model with the least molecular clashes (see Methods). It should be noted that the number of clashes is relatively high for the best models of PKM Apl I and KIBRA_{WT} (Fig. 3; green path 0.267 Å, orange alt path 0.434 Å). The best model for PKM Apl I and KIBRA_{RSR-AAA} when SP3 is in the substrate site had fewer clashes (0.208 Å; pLDDT = 66.8, iTM = 0.76) as did the best model when SP1 is in the substrate site (0.153 Å; pLDDT = 58.2, ipTM = 0.6). In contrast, Molprobit shows no strong model (mean clash pre residue < 0.9 Å and 100% favored Ramachandran scores) for PKM Apl III and KIBRA_{RSR-AAA} with SP3 in the binding pocket and the number of clashes is high with SP1 (0.414 Å; pLDDT = 53.1, ipTM = 0.71).

A positively charged alpha-helix in KIBRA that interacts with different regions of PKM Apl I and PKM Apl III gains negative charges after splicing

KIBRA_{splice} stabilizes PKM Apl I, but not PKM Apl III, but KIBRA_{splice} shows strong BIFC with PKM Apl III. The splice occurs in a region of KIBRA after the SPs that AlphaFold 3 predicts consists of four alpha helices (Fig. 5A). The first of the helices is predicted to form the second coiled-coil (CC) domain with one of the helices from the N-terminal region. The splice occurs between the second and third predicted alpha helices. All the sequences predicted to form these helices are highly conserved [10] (Fig. 5A), and either are predicted to be part of or sandwiched in between the three CC domains (Fig. 5B). These interactions and helices are also predicted for vertebrate KIBRA (WWC1), WWC2 and WWC3 (Supplementary Fig. 1). In particular, the third of these helices is predicted to have a face with mainly positive charged amino acid side chains. Some of these positively charged residues are predicted to interact with the region of PKM Apl III predicted to interact with the inhibitory arm of the KIBRA substrate peptide (Fig. 5C). The splice added seven amino acids to the linker between two of these helices (Fig. 5B) and importantly, is predicted to allow negatively charged residues previously in the linker region between the second and third predicted alpha-helix to join the face of the helix (Fig. 5D). The pLDDT measure of AlphaFold3 folding confidence of the splice, linker, and positive faced helix was similar with KIBRA_{WT} 63.38 and KIBRA_{splice} 63.07 (residues 1406–1439 for KIBRA_{WT} and 1406–1446 for KIBRA_{splice}), suggesting the difference between structures is not a result of a change in folding confidence but

rather a potential real structural change that may underlie the observed differences with KIBRA_{splice}. Consistent with this structural change underlying the difference with KIBRA_{splice}, the AlphaFold 3 models with KIBRA_{splice} are predicted to have reduced interactions between the positively charged face of the helix and PKM Apl III (Fig. 5E), also apparent as a change in the mean minimum iPAE at the helix (from 14.8 ± 1.0 to 17.6 ± 2.1 Å, for PKM Apl III with positive faced helix of KIBRA_{WT} (1419–1440) and KIBRA_{splice} (1419–1446) respectively, mean over 25 models $\pm$ pooled SD). Some of the negative charges important for the predicted interaction of the positively charged helix with PKM Apl III are not conserved in PKM Apl I (Fig. 5E). In particular, an arginine (gray) at position 185 in PKM Apl I is predicted to strongly decrease the interaction (Fig. 5E). Instead, for PKM Apl I the helix is predicted to interact with a group of negative charged side chains in the handle domain, a region not conserved in PKM Apl III (Fig. 5F). Some of these interactions are modelled to reduce in KIBRA_{splice}, particularly with the two acidic residues closest to the alternative – 7R binding pocket (Fig. 5F). The ‘affinity arm’ interacting FxR of KIBRA SP3 is often modelled interacting with the PKM Apl I handle (Fig. 3) while either SP1 or SP2 is predicted to interact with the affinity arm-interacting region of PKM Apl I, something that is not observed in models with PKM Apl III. The occurrence of this predicted alternate interaction of the canonical substrate peptide SP3 with the PKM Apl I handle domain is reduced in models with KIBRA_{splice} (5 of the 25 models, compared to 12 of 25 models with KIBRA_{WT}). The AlphaFold models of KIBRA_{splice} with PKM Apl I also have reduced interactions with the positively charged alpha helix and the handle domain, particularly with the two acidic residues E230 and E234 (Fig. 5F inset chart), the closest acidic residues to D228 which is modelled to be a key interaction supporting the alternate – 7R binding site in the – 2R pocket of PKM Apl I (Fig. 5Ffar right). The mean minimum iPAE of PKM Apl I for SP3 is decreased (predicted interaction increased) with KIBRA_{splice} as it is predicted to be more likely that KIBRA SP3 follows the path that is similar to that predicted of PKM Apl III (Fig. 5G). This is correlated with increased stabilization of PKM Apl I by KIBRA_{splice} [10].

Effects of KIBRA variants on stabilization and BIFC

Several mutations in WWC3 in the substrate binding and helices region cause neurodevelopmental disorders in humans or transmissible cancer in vertebrates [18, 19, 40]. To determine the importance of these mutations we converted the orthologous residues in *Aplysia* KIBRA and KIBRA_{splice} to the same amino acid found in these cases (arginine at residue 1312 to glutamine; KIBRA_{R-Q} and arginine at residue 1341 to histidine KIBRA_{R-H}, arginine

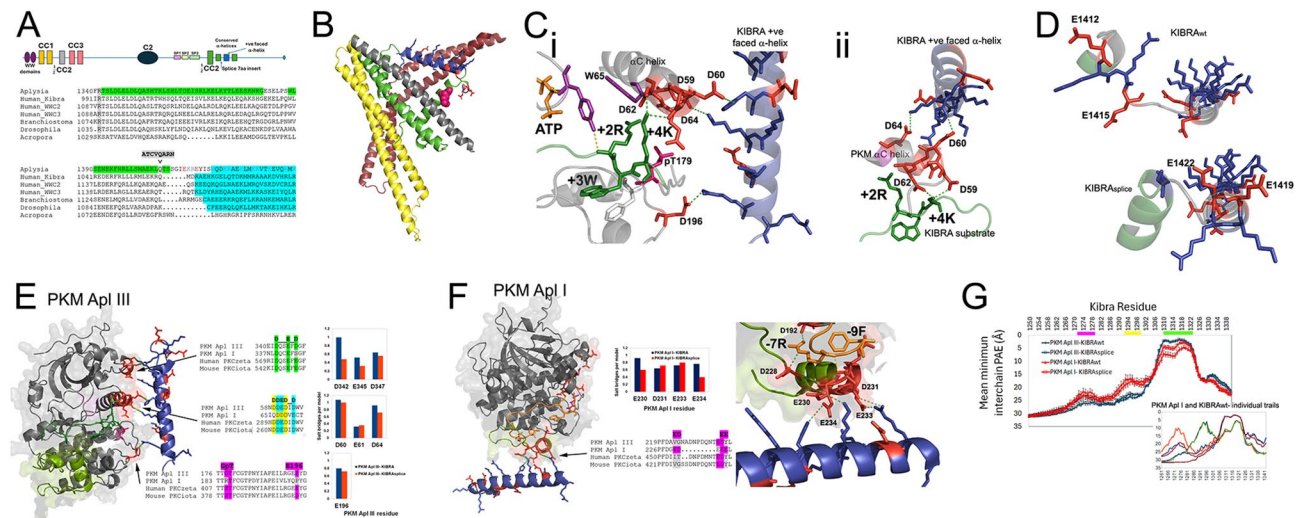


Fig. 5 Splice insert differentially alters interaction of positively charged α -helix with PKM Apl III and PKM Apl I. **A** Domain map of *Aplysia* KIBRA and amino acid sequence alignment of KIBRA from *Aplysia*, *Branchiostoma*, *Drosophila*, and *Acropora*, and the three Human orthologs, KIBRA, WWC2, and WWC3. The R-H mutation is marked as are the locations of three conserved post-PKC substrate helices, the first is the other half of the second coiled coil domain (green in B), the second α -helix has the splice site marked in grey with the insert noted above, and the third α -helix highlighted in cyan has a face of positive charged side chains R, K, H, in blue in *Aplysia*. The two residues with acidic side chains that enter the positive face of the helix with splice in red. **B** Representative cartoon structure generated by AlphaFold 3 of *Aplysia* KIBRA colored according to A, with CC1 in yellow, CC3 in red, CC2 grey/green. Only CC regions and post-substrate alpha helices are visible with the post-substrate helices in green and the positive faced helix in blue with the splice site marked with pink spheres. Residue side chains visible and colored according to charge for K, R, H, E, D from the splice site through the positive faced helix. **C** i. AlphaFold 3 models of the inhibitory arm of KIBRA_{WT} SP3 (+2RWK) interacting with PKM Apl III as in Fig. 3Dii, now also showing potential concurrent interactions from the positive faced helix that tends to be modeled salt bridging to the other two aspartic acid residues in the α C helix acidic patch (D60, D64). 22 of 25 models of PKM Apl III with KIBRA_{WT} predict an interaction with D60 and D64 with the positive-faced helix. ii) Different viewing angle of i, with just the side chains of the residues participating in salt bridges formed with the α C helix visible. This is an example, specific residues interacting from the positive faced helix vary model to model, reducing the average iPAE of this interaction. **D** The positive faced helix with all charged side chains visible has nine residues with positive charged side chains facing one direction and E1412&E1415 side chains facing away (top). In all 25 models, including the 16 models without Ramachandran outliers in this region, E1412&E1415 are modelled to face away from the positive faced helix in KIBRA_{WT}. The splice site with the 7 amino acid insert (green) in KIBRA_{splice}, the negative charged E1412&E1415, now E1419&E1422 are predicted by AlphaFold to participate in the helix with the negative side chains in the same direction as the positive face of the helix (bottom). This modelling of E1419&E1422 into the positive face in KIBRA_{splice} occurs in 11 of 14 models without Ramachandran outliers over this region. **E** Representative AlphaFold 3 generated model of PKM Apl III with KIBRA, only SP3 (green) and the positive faced helix visible from KIBRA (K, R, H and D, E side chain sticks visible, blue and red respectively). The positive faced helix interacts with three regions in PKM Apl III, D342, D345, D347 (green), E196, G178, pT179 (magenta), and D60, E61, D64 in the negative charged patch in inhibitory arm interacting region (cyan). Sequence alignments highlight the similarities between atypical PKC/PKM and PKM Apl I, coloring shows conservation, with grey highlighting differences in PKM Apl I and yellow showing the residues interacting the KIBRA substrate at the inhibitory arm interacting region. **F** Representative model of the alternative interaction with PKM Apl I and KIBRA_{WT}, only SP3 (orange) and the positive faced helix visible from KIBRA. The positive faced helix interacts with a single region in the handle of PKM Apl I (yellow) at E230, D231, E233, E234 (red), where the interactions with E230 and E234 decrease in KIBRA_{splice} models. (Charts in E&F) Mean predicted number of salt bridges formed between the KIBRA positive faced α -helix and specific residues in the PKMs over twenty-five models for each KIBRA_{WT} and KIBRA_{splice}, normalized per model. With PKM Apl III, the positive faced helix interacts consistently with three locations, two of which have a reduced number of interactions when modelled with KIBRA_{splice}. With PKM Apl I the positive faced helix interacts with a single region in the handle and this interaction is reduced with two of the residues E230 and D234 in KIBRA_{splice} models, compared to KIBRA_{WT}. Right, closeup of the patch of acidic side chains (red) in the PKM Apl I handle (olive) that participate in the alternate SP3 binding (-7R orange) and interaction with the positive faced helix of KIBRA_{WT} (grey, with positive charged side chains in blue). **G** Mean minimum iPAE for the KIBRA substrate region of models with KIBRA and KIBRA_{splice} with either PKM Apl III or PKM Apl I. Representative individual models in inset show variability in minimum iPAE between models with PKM Apl I and KIBRA that lead to increased mean minimum iPAE at SP1 and SP2. Specific representative models are the same used in Fig. 3 and the model for KIBRA_{splice} with PKM Apl III for 5D had a substrate interface mean clash per residue 0.035Å, substrate interface pLDDT=73.15, and ipTM=0.87

at residue 1312 to a stop codon KIBRA_{stop}; Figs. 4A and 5A). R1312 is the R in FxR affinity arm and R1341 is the first residue in the first alpha helix that participates in the CC domains with the N-terminal region of KIBRA (Fig. 5B).

For stabilization assays, KIBRAs were generated as fusion proteins with eCFP at the N-terminus to allow comparison of expression levels. The level of mRFP-PKM

Apl III was increased by eCFP-KIBRA_{WT}, eCFP-KIBRA_{R-H} and eCFP-KIBRA_{R-Q} when standardized to the eCFP alone group (Fig. 6A; quantified in Fig. 6B). Surprisingly, expression of CFP-KIBRA_{stop} in which the proposed PKM binding site and the helices that contribute to the coiled-coil region in the C-terminal is removed still stabilized PKM Apl III (Fig. 6A; quantified in Fig. 6B). All these mutants were expressed at higher levels than

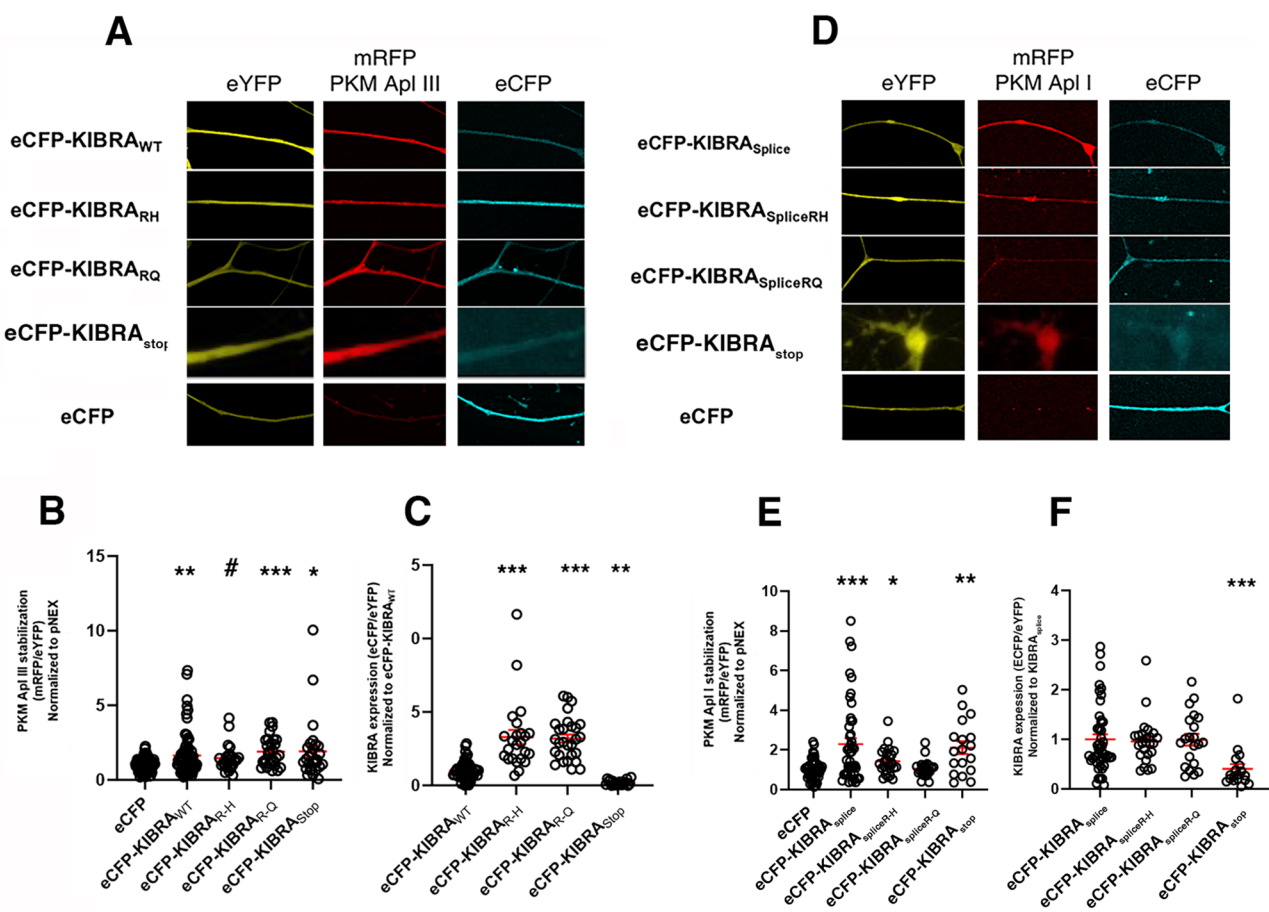


Fig. 6 Stabilization of KIBRA variants. **A** Representative examples of expression of eYFP (control for injection) eCFP (eCFP or eCFP-tagged KIBRA) and mRFP (PKM Apl III) for the five groups (eCFP, eCFP-KIBRA_{WT}, eCFP-KIBRA_{RH}, eCFP-KIBRA_{RQ}, eCFP-KIBRA_{stop}) are shown. **B** All individual points (black circles) and the average and SEM (red) of the normalized mRFP/eYFP ratio (Stabilization, see Methods) is shown. Kruskal-Wallis test (253,5) = 26.28, $p < 0.0001$. Dunn's multiple comparison test showed that three of the groups were significantly different than eCFP ($p < 0.001$, ***; $p < 0.01$, **; $p < 0.05$, *). The eCFP-KIBRA_{RH} group was only of borderline significance (#, $p = 0.08$). The N was the number of sensory neurons quantified (eCFP 89, eCFP-KIBRA_{WT} 84, eCFP-KIBRA_{RH} 24, eCFP-KIBRA_{RQ} 28, eCFP-KIBRA_{stop} 28) from 10 independent experiments. **C** All individual points (black circles) and the average and SEM (red) of the normalized eCFP/eYFP ratio (Expression, see methods) is shown. Kruskal-Wallis test (125,4) = 85.83, $p < 0.0001$. Dunn's multiple comparison test showed that all of the groups were significantly different than eCFP-KIBRA_{WT} ($p < 0.001$, ***; $p < 0.01$, **). The N and independent experiments is the same as for (B). **D** Representative examples of expression of eYFP (control for injection) eCFP (eCFP or eCFP-tagged KIBRA) and mRFP (PKM Apl I) for the five groups (eCFP, eCFP-KIBRA_{splice}, eCFP-KIBRA_{spliceRH}, eCFP-KIBRA_{spliceRQ}, eCFP-KIBRA_{stop}) are shown. **E** All individual points (black circles) and the average and SEM (red) of the normalized mRFP/eYFP ratio is shown. Kruskal-Wallis test (159,5) = 21.84, $p < 0.001$. Dunn's multiple comparison tests showed that eCFP-KIBRA_{splice} ($p < 0.001$, ***), eCFP-KIBRA_{spliceRH} (*, $p < 0.05$) and eCFP-KIBRA_{stop} (**, $p < 0.01$) were significantly different than eCFP. In contrast, eCFP-KIBRA_{spliceRQ} was not significantly different than eCFP ($p > 0.5$). The N was the number of sensory neurons quantified (eCFP 48, eCFP-KIBRA_{splice} 46, eCFP-KIBRA_{spliceRH} 25, eCFP-KIBRA_{spliceRQ} 22, eCFP-KIBRA_{stop} 18) from 8 independent experiments. **F** All individual points (black circles) and the average and SEM (red) of the normalized eCFP/eYFP ratio (Expression, see methods) is shown. Kruskal-Wallis test (110,4) = 22.25, $p < 0.0001$. Dunn's multiple comparison test showed that only eCFP-KIBRA_{stop} was significantly different than eCFP-KIBRA_{WT} ($p < 0.001$, ***; The N and independent experiments is the same as for E).

CFP-KIBRA_{WT} (Fig. 6C), except for CFP-KIBRA_{stop} that was expressed at significantly lower levels (Fig. 6C). For PKM Apl I stabilization all mutations were made in the CFP-KIBRA_{splice} background as PKM Apl I is stabilized by KIBRA_{splice} but not KIBRA_{WT}. The stop codon in KIBRA_{stop} precedes the splice so this construct was the same in both assays. In contrast to PKM Apl III, CFP-KIBRA_{spliceRQ} did not stabilize PKM Apl I, while CFP-KIBRA_{splice}, CFP-KIBRA_{spliceRH} and CFP-KIBRA_{stop} all stabilized PKM Apl I (Fig. 6D, quantified in Fig. 6E). All

CFP-KIBRAs expressed at similar levels, except for CFP-KIBRA_{stop} that was expressed at significantly lower levels (Fig. 6F), consistent with experiment examining stabilization of PKM Apl III (Fig. 6C). AlphaFold3 predictions of the structure of KIBRA_{stop} shows a lack of formation of the coiled coil structures seen in KIBRA_{WT} (Supplemental Fig. 2) and this may lead to decreased stability of this protein.

We also compared BIFC levels for KIBRA variants and PKM Apl III. Significant BIFC was seen for all constructs,

including KIBRA_{stop} (Fig. 7A, quantified in Fig. 7B). All constructs were expressed at equal levels (Fig. 7C), but since the antibody was generated to a C-terminal peptide, it did not recognize KIBRA_{stop} and we assume that it expressed at lower levels than the other constructs based on levels seen with eCFP-KIBRA_{stop} (Fig. 6).

We examined the predicted effect of these mutations on the AlphaFold 3 models. There were no major changes predicted with the KIBRA_{R-Q} and KIBRA_{R-H} mutants on PKM Apl III (Fig. 8A), although there was a slight decrease in iPAE for SP1. In contrast, KIBRA_{spliceR-Q} had larger predicted changes in the relative binding of the SPs for PKM Apl I (Fig. 8B) consistent with a larger effect on stabilization for this change. As would be expected given the loss of SP3, KIBRA_{stop} is predicted to show enhanced binding of SP1 and SP2 to the PKMs (Fig. 8C). While the iPAE values for SP1 and SP2 for KIBRA_{stop} appear to show a weaker predicted interaction than seen for SP3 for KIBRA, this is somewhat misleading as it reflects models with either SP1 or SP2. When considering individual

models, low iPAE values indicative of a strong interaction are modelled with KIBRA_{stop} (25 PKM Apl III models show peak minimum iPAE of 4.6 ± 0.8). PKM Apl III is modelled as slightly more biased to SP2 and PKM Apl I slightly biased to SP1 (Fig. 8C). This is consistent with the prediction that either SP1 or SP2 alone in interaction with PKMs can lead to stabilization. SP1 and SP2 both start with the affinity arm interacting Ψ xR, IxR and LxR, and AlphaFold 3 models suggest these residues interact similarly as SP3 with the PKMs (ipTM 0.69 ± 0.05 and 0.64 ± 0.05 for KIBRA_{stop} models with PKM Apl III that have preference to SP1 (10) or SP2 (15), and 0.62 ± 0.02 and 0.74 ± 0.03 for KIBRA_{stop} models with PKM Apl I with preference to SP1 (18) and SP2 (6) with one model not showing a confident substrate interaction). The mean clash per residue is low for the KIBRA_{stop} alternate substrates, as low as $0.189\text{\AA}$ and $0.112\text{\AA}$ with PKM Apl III and as low as $0.066\text{\AA}$ and $0.063\text{\AA}$ with PKM Apl I for models preferring SP1 and SP2 respectively (data from models selected with the same criteria as for SP3,

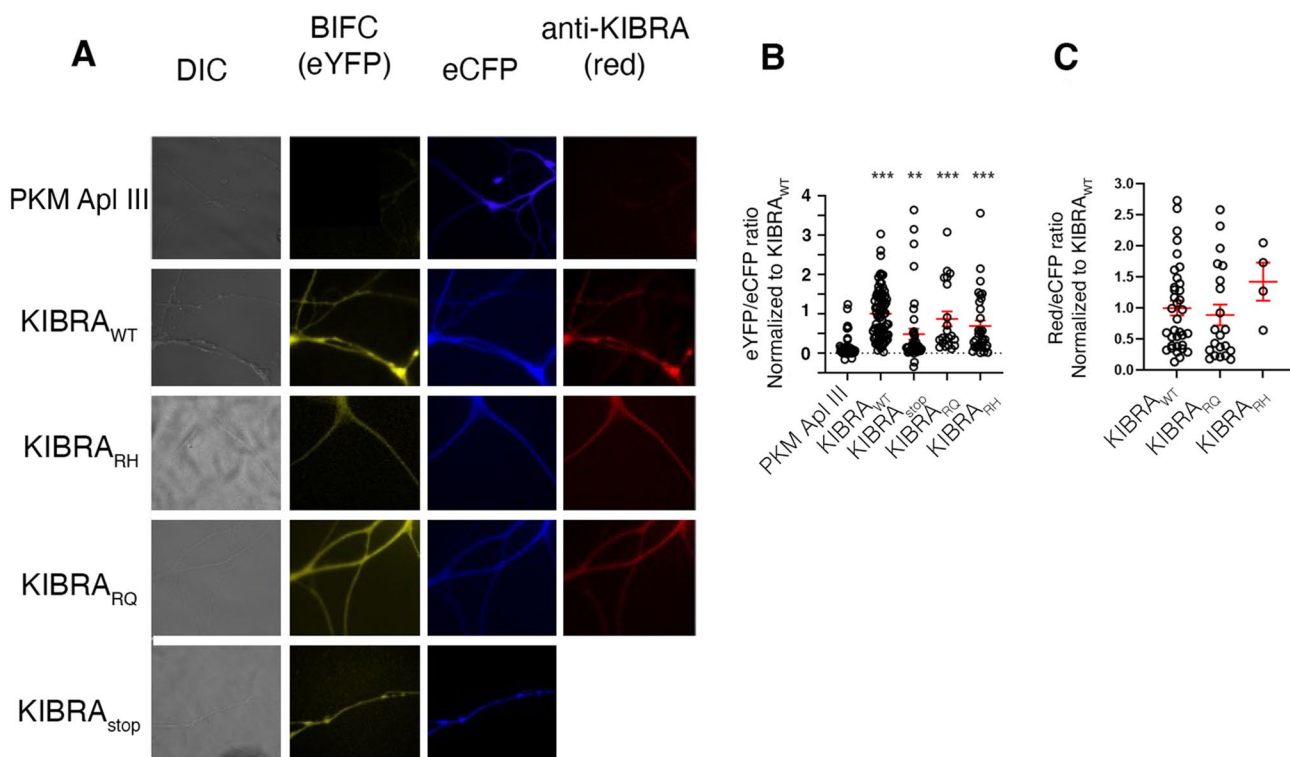


Fig. 7 BIFC of KIBRA variants. **A** Representative examples of expression of DIC (Differential Interference Contrast), eYFP (BIFC), Red (Kibra Antibody), and eCFP (control for injection) are seen for the groups all expressing PKM Apl III short and one of: PKM Apl III long, KIBRA_{WT} Long, KIBRA_{R-Q} Long, KIBRA_{R-H} Long, KIBRA_{stop} Long **B** All individual points (black circles) and the average and SEM (red) of the normalized eYFP/eCFP ratio (BIFC, See methods). Kruskal-Wallis test (229,5)=97.15 showed a significant difference between groups $p < 0.0001$. Dunn's multiple comparison tests showed that all groups were significantly different that PKM Apl III long ($p < 0.001$, ***, $p < 0.01$, **). N was the number of sensory neurons quantified: PKM Long 52, KIBRA_{WT} Long 82, KIBRA_{R-Q} Long 41, KIBRA_{R-H} Long 33, KIBRA_{stop} Long 41). **C** All individual points (black circles) and the average and SEM (red) of the normalized red/eCFP ratio are shown. Kruskal-Wallis test (72,4)=31.69 showed a significant difference between groups. Dunn's multiple comparison tests showed that all groups were significantly different than PKM Long ($p < 0.001$, ***, $p < 0.01$, **). KIBRA_{R-Q} Long and KIBRA_{R-H} Long were not expressed differently than KIBRA_{WT} Long ($p > 0.5$). N was the number of sensory neurons quantified and different from (B) as not all experiments had antibody staining (PKM Long 12, KIBRA_{WT} Long 35, KIBRA_{R-Q} Long 21, KIBRA_{R-H} Long 4). The antibody does not recognize KIBRA_{stop} as the epitope used to raise the antibody is at the carboxy-terminal

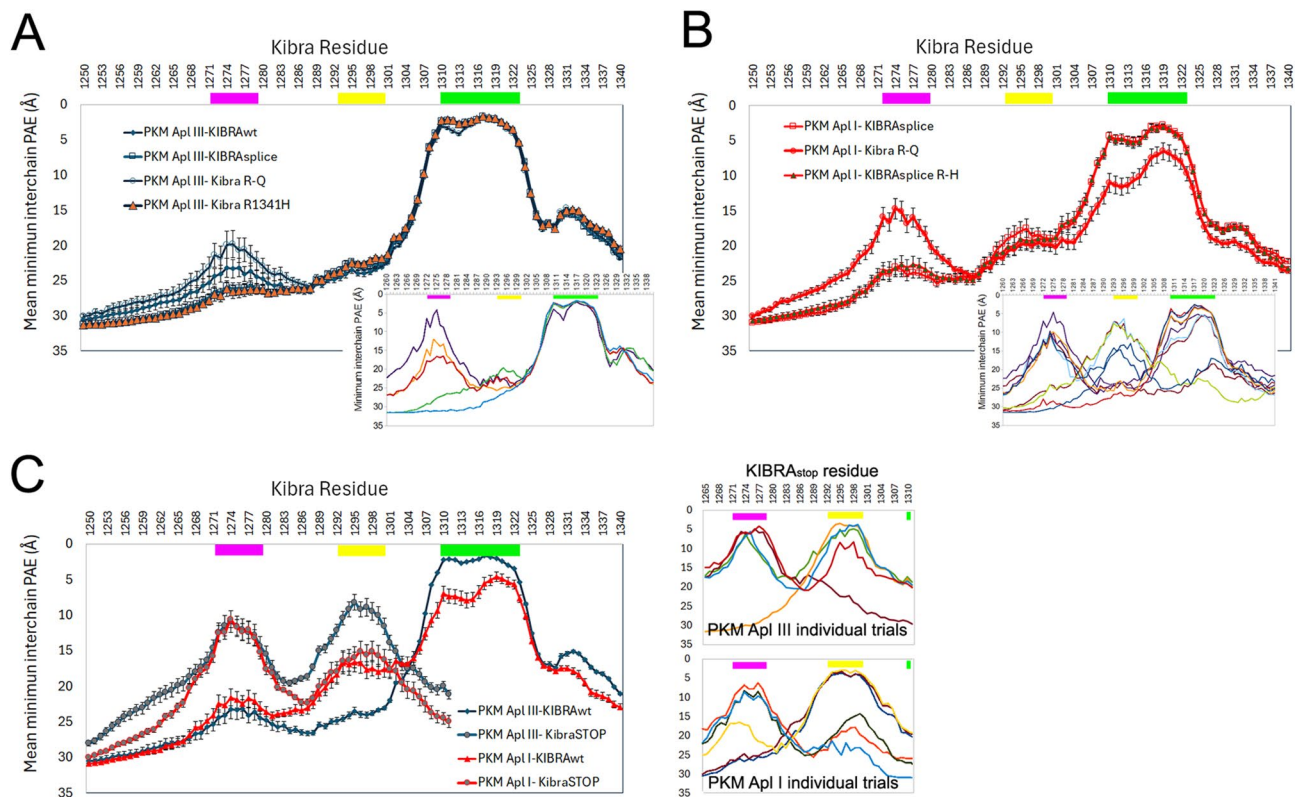


Fig. 8 AlphaFold 3 models of KIBRA variants. Mean minimum interchain PAE for KIBRA substrate region for KIBRA mutants, KIBRA R-Q (1312) and KIBRA R-H (1341) with PKM Apl III (**A**) or PKM Apl I (**B**) (measured from 25 models in each group). Insets show representative individual AlphaFold model minimum iPAE measurements revealing variation in interaction sites between individual models. **C** Mean minimum interchain PAE for KIBRA_{stop} with PKM Apl III and PKM Apl I, with representative individual trials on the right.

no severe clash (>0.9 Å) 100% Ramachandran favored residues).

Discussion

Overall, our results highlight the role of predicted substrate-like peptides in KIBRA for AlphaFold 3 predicted KIBRA-PKM interactions. The AlphaFold 3 models show differences in this interaction with KIBRA variants, and between PKM Apl I and PKM Apl III. Examining the iPAEs in the SP1, SP2 and SP3 regions together with examining of the clashes predicted by Molprobit for the models [47] provide hypotheses to explain the results we observe. The change in clashes with KIBRA_{RSR-AAA} for PKM Apl I (decreased clashes) to PKM Apl III (increased clashes) is consistent with switches in the stabilization of PKM Apl I and PKM Apl III by KIBRA_{RSR-AAA}. Similarly, the decreased clashes in KIBRA_{stop} for both PKM Apl I and PKM Apl III are consistent with the stabilization of both isoforms by this construct. Alpha Fold 3 predictions partially explained the differential effects of the splice on PKM Apl I and PKM Apl III. It predicts the insertion of negative changes into the alpha helix to change its interactions, and since the alpha helix affected by the splice is predicted to interact very differently with PKM Apl I

and PKM Apl III suggests reasons the splice affects the two interactions distinctly. However, the reasons for the differential stabilization of PKM Apl I and PKM Apl III by KIBRA_{splice} is not explained by these differences. We suggest that a critical feature that is difficult to predict from the models is whether the substrates will actually be phosphorylated or the binding leads to strong inhibition as experimentally shown for the PAR3 and KIBRA peptides for the atypical PKC iota [38]. Thus, we would predict that KIBRA_{splice} no longer stabilizes PKM Apl III because alterations in the interaction between the inhibitory arm of the KIBRA peptide with PKM Apl III due to splicing leads to increased phosphorylation of the KIBRA peptide. It should be noted that even the PAR3 peptide in isolation shows some phosphorylation by atypical PKCs using more sensitive assays [13]. We would predict the opposite effect for PKM Apl I, stabilization of an inhibitory state after the splice.

The possible role of KIBRA as an inhibitor of PKCs/PKMs is consistent with some models [51], but not others [45, 44]. Like the PAR3 peptide, and unlike a pseudosubstrate, there is a phosphorylatable residue in the KIBRA SPs. In the polarity pathway PAR3 stabilizes atypical PKC in an apical complex where the kinase is inhibited by this

peptide. Activation of PKC phosphorylation of PAR3 breaks up the complex leading to PAR3 re-localizing to junctions and PKC phosphorylation of key substrates in organizing cell polarity [38]. Similarly, atypical PKC binds to its substrate LGI in a stalled configuration with binding to cdc-42 activating phosphorylation and dissociation [7]. We would suggest that regulated phosphorylation of SPs in the KIBRA-PKM interaction regulates the function of this complex.

The simplest interpretation of the discrepancies we observe between BIFC and stabilization is to assume that BIFC interactions are more sensitive to transient events than stabilization. This is not surprising, since with some probability transient interactions will lead to joining of the two parts of the fluorescent proteins and once joined, this interaction may no longer require continuous PKM-KIBRA interactions. In contrast, we envision that destabilization of PKMs occur when PKMs are not associated with KIBRA and that leads to degradation of the kinase. Once the protein is degraded, it can no longer be protected by KIBRA. Thus, we expect to see association with BIFC with semi-stable interactions, but not stabilization since semi-stable interactions would allow for degradation. This model is consistent with the data we observe, seeing BIFC without stabilization, but not the reverse.

We had shown previously that isoform specific stabilization of PKMs by KIBRA depended on differences in the “handle” domain [10]. Similar to PKM Apl I, PKM Apl III containing the PKM Apl I handle is stabilized by KIBRA_{splice} but not KIBRA. Moreover, small molecules that bind to the handle domain act as PKC inhibitors and reduce PKMζ BIFC with KIBRA [44]. Thus, we expected that AlphaFold 3 would predict direct interactions between the handle domains and KIBRA. We do observe predicted interactions between the handle domain of PKM Apl I, both with the affinity arm of the substrate peptide and the alpha helix following the splice and these interactions are consistent with the effects of stabilization when the PKM Apl I handle replaces the PKM Apl III handle. In contrast, the PKM Apl III handle domain is predicted to show quite weak interaction with KIBRA and is the only region of PKM Apl III that shows high iPAE in the KIBRA-PKM AlphaFold 3 model (Supplemental Fig. 3). This is conserved in the predicted PKMζ-KIBRA interaction (Supplemental Figure 3). We have not examined how the atypical PKM inhibitors targeted to the handle domain [44] affect BIFC in *Aplysia* since neither of these inhibitors is expected to bind to PKM Apl III as they are targeted to residues that are not conserved in the *Aplysia* kinase. In a recent paper, the handle domain of PKMζ is predicted to stabilize a hexameric state of PKMζ-KIBRA by interacting between the heterodimers [14]. This region is free in our dimeric

structure (Supplementary Fig. 3) and the arginines providing the hydrogen bonds in this model are conserved.

Limitations of the study

Our results in *Aplysia* sensory neurons are all based on overexpression of KIBRA and PKMs. While this allows for examination of interaction and stabilization in a neuronal context, the level of overexpression is quite high. While we cannot do immunoblots after expression in individual neurons, the level of immunoreactive KIBRA and PKMs compared to non-expressing cells is quite large ([10]Fig. 2). This is important as it is likely that endogenous complexes of KIBRA and PKM contain additional proteins that will affect the interactions we observe but are unlikely to be equimolar with the high concentrations of KIBRA and PKM in these studies. For example, PICK can interact with both KIBRA [29, 37] and PKMs [46, 48]. Similarly, proteins have been identified that bind to the C2, WW and PDZ ligand of KIBRA and these may lead to distinct complexes [12, 35, 41]. Similarly, ligands can bind to the handle domain of PKMs and may compete with KIBRA peptides [27]. PS1 and PS2 have been shown to be RSK substrates [50] and this suggests RSK may be an important regulator of KIBRA-PKM interactions. We believe these additional interactions are necessary for a model where KIBRA-PKM interactions are important for the maintenance of memory, as in this model KIBRA localizes/stabilizes an active form of PKM, while the predicted interactions with KIBRA peptides occupying the substrate binding domain of PKM would act as inhibitors.

Conclusions

Our experiments demonstrate that BIFC does not predict stabilization for KIBRA-PKM interactions and highlight the importance of KIBRA SPs interacting with PKMs for isolated PKM-KIBRA interactions. We also highlight the possible importance of the highly conserved alpha helices following the KIBRA-PKM interacting peptide in controlling KIBRA-PKM interactions. We would suggest that this inhibitory interaction between KIBRA SPs and PKMs are important at distinguishing synapses not involved in memory and synapses that depend on persistent PKM activity. Activation of phosphorylation of the SPs would free PKM to phosphorylate the substrates important for retaining memory, such as putative endocytic adaptors [9, 31, 34], similar to the role of the PAR3-atypical PKC complex in regulating cell polarity [38]. Understanding the transition from an inhibitory interaction to one where KIBRA complexes with an active PKM is a key question for future studies.

Abbreviations

BIFC	Bimolecular fluorescence complementation
CC	Coiled-coil

CFP	Cyan fluorescent protein
GFP	Green fluorescent protein
iPAE	Interchain predicted alignment error
KIBRA	Kidney brain protein
LATS	Large tumor suppressor kinase
LTF	Long-term facilitation
PAR3	Partitioning defective 3
PKC	Protein kinase C
PKM	Protein kinase M
mRFP	Monomeric red fluorescent protein
ROI	Region of interest
SP	Substrate peptide
WWC	WW and C2 domain containing
eYFP	Enhanced yellow fluorescent protein

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13041-026-01321-2>.

Supplementary Material 1

Acknowledgement

Not applicable.

Author contributions

EA, TD and LF performed experiments, analyzed data and participated in writing and editing the manuscript, KBD and CO performed experiments and analyzed data. JK and WS conceived the study. WS analyzed data and participated in writing and editing the manuscript.

Funding

This work was supported by CIHR Project grant 180274.

Data availability

All custom constructs and antibodies are freely available on request from the Sossin lab. The raw data used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

All animal experiments were carried out in compliance with protocols approved by the Montreal Neurologic Institute animal committee (Protocol MNI-5784).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 4 March 2026 / Accepted: 1 June 2026

Published online: 19 June 2026

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